25900749 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCBTG100, MG1DSCBBT101 - FUNDAMENTALS FOR BIOTECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Answer the Questions in One Word or Sentence

Answer **All** Questions

Each Question carries **1** mark

1. Name the type of cell that synthesizes immunoglobulin. [U] / [CO3]
2. Which enzyme unwinds the DNA helix at the replication fork? [U] / [CO2]
3. Give an example of conventional iotechnology. [K] / [CO1, CO2]
4. Name two industries where biotechnology is applied. [K] / [CO1, CO2]
5. Who contributed to the development of cell theory alongside Mathias Jakob Schleiden. [K] / [CO1, CO2]
6. State the different types of nitrogen bases. [K] / [CO1, CO2]
7. who is the father of rDNA technology [U] / [CO2]
8. What is the main reproductive structure of fungi? [U] / [CO1, CO2]
9. What type of RNA carries genetic information? [K] / [CO2]
10. Which company was the first to commercialize Humulin? [K] / [CO1, CO2]

[10x1 = 10]

Part B

Short Answer Questions

Answer any **4** Questions

Each Question carries **3** marks

11. Give an outline of major milestones in biotechnology. [K] / [CO1, CO2]

12. How does the secondary structure of a protein form? [K] / [CO1, CO2]

13. Explain monohybrid and dihybrid crosses with examples. [U] / [CO3]

14. Describe the solubility properties of carbohydrates in water and explain their significance. [U] / [CO1, CO2]

15. State the function of golgi apparatus? [K] / [CO1, CO2]

16. Describe the components in the structure of a virus. [K] / [CO1, CO2]

[4x3 = 12]

Part C

Short Essay Questions

Answer any **3** Questions

Each Question carries **6** marks

17. Summarize the structures and functions of prokaryotic and eukaryotic cells, highlighting the key differences. [U] / [CO1, CO2]

18. Explain the action of recombinant vaccines in COVID-19. [U] / [CO2]

19. What is the role of mRNA in translation? [U] / [CO2]

20. explain the history of biotechnology? [U] / [CO1, CO2]

21. State the three fundamental laws of inheritance proposed by Gregor Mendel. [K] / [CO3]

[3x6 = 18]

Part D

Essay Questions

Answer any **1** Question

Each Question carries **10** marks

22. Outline the composition and functions of phospholipids in biological membranes. [K] / [CO1, CO2]

23. Determine the important cells involved in immune system. [An] / [CO3]

[1x10 = 10]